

PG-Sheets - KG20C Scholarly Knowledge Graph

EXECUTIVE SUMMARY

Dataset Name	KG20C Scholarly Knowledge Graph
Number of Node Labels	5
Number of Edge Labels	5
Total number of nodes	16362
Total number of edges	55607
Version	Not informed
Source	Kaggle and Author's Github ¹²
Dataset authors	Hung Nghiep Tran and Atsuhiko Takasu
How to cite	See bibtex reference bellow
PG-Sheets authors	Larissa C. Shimomura

To cite the dataset, cite the original paper in which this dataset was used, the bibtex to use for this citation follows:

```
@inproceedings{DBLP:conf/ercimdl/TranT19,  
  author      = {Hung Nghiep Tran and  
                 Atsuhiko Takasu},  
  editor      = {Antoine Doucet and  
                 Antoine Isaac and  
                 Koraljka Golub and  
                 Trond Aalberg and  
                 Adam Jatowt},  
  title       = {Exploring Scholarly Data by Semantic Query on Knowledge Graph Embedding  
                 Space},  
  booktitle   = {Digital Libraries for Open Knowledge - 23rd International Conference  
                 on Theory and Practice of Digital Libraries, {TPDL} 2019, Oslo, Norway,  
                 September 9-12, 2019, Proceedings},  
  series      = {Lecture Notes in Computer Science},  
  volume      = {11799},  
  pages       = {154--162},  
  publisher   = {Springer},  
  year        = {2019},  
  url         = {https://doi.org/10.1007/978-3-030-30760-8\_14},  
  doi         = {10.1007/978-3-030-30760-8\_14},  
  timestamp   = {Mon, 26 Jun 2023 20:46:07 +0200},  
  biburl      = {https://dblp.org/rec/conf/ercimdl/TranT19.bib},  
  bibsource   = {dblp computer science bibliography, https://dblp.org}  
}
```

Disclaimer: This PG-Sheets was written by Larissa C. Shimomura, however, she did not participate in the construction and publication of the dataset. All information written in this PG-Sheets is written based on the publications of the original authors, and all credits on the datasets should be given properly according to the dataset license.

COMPOSITION (WHAT?)

Context/Domain of the data - The dataset comes from the digital libraries domain in specific, this dataset is a citation network between published papers in top venues.

Information about the instances Figure 1 contains the schema of the dataset, labels, and attributes of each node of edge. The Figure 1 is a figure create by Larissa based on the author's figure³. All nodes have an id,

³https://github.com/tranhungnghiep/KG20C/blob/main/KG20C_graph.png

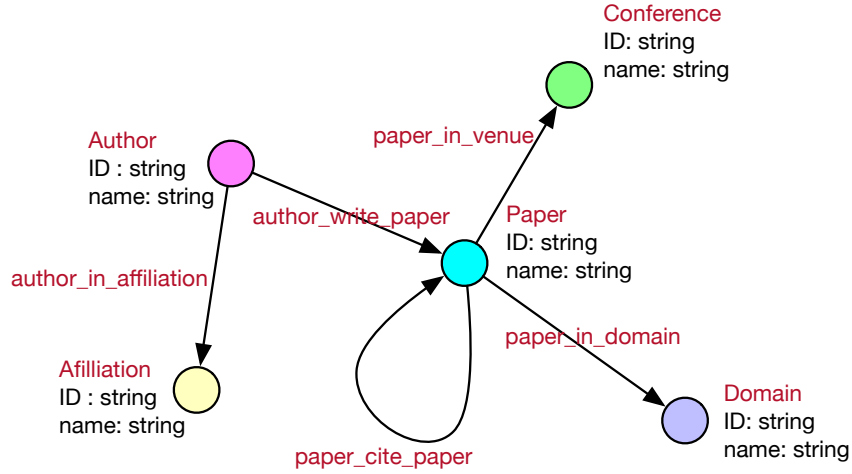


Figure 1: KG20C Schema

name and a label also referred to as type in the dataset Kaggle webpage.

Label	Attribute/Property	Description
Author	ID	Identifier of the authors entity
Author	Name	Name of an author as it appears in the paper title
Paper	ID	Identifier of the paper in this dataset, does not refer to DOI or other unique identifiers
Paper	Name	Title of the paper
Conference	ID	Identifier of the conference
Conference	Name	Name of the conference
Domain	ID	Identifier of the domain entity
Domain	Name	Research domain or topic (e.g. Artificial Intelligence)
Affiliation	ID	Identifier of the affiliation nodes
Affiliation	Name	Name of university, company or insitute or companies in which the authors are affiliated

Table 1: Table with description of the properties

Node Label	Size
Author	8680
Paper	5047
Conference	20
Domain	1923
Affiliation	692

Edge Label	Size
author_in_affiliation	Number of edges
author_write_paper	
paper_cite_paper	
paper_in_domain	
paper_in_venue	

The number of edges are the total number in the training and test sets of the graph. In the following, we show the number of edges in each of the sets provided.

Edge Label	Train	Test
author_in_affiliation	Number of edges	
author_write_paper		
paper_cite_paper		
paper_in_domain		
paper_in_venue		

Table 2: Number of edges per set

Important additional information - In this section we describe some additional information about the dataset and their attributes.

- All IDs are unique, independent of their labels. That means that there is one unique ID for each one of

the nodes.

- As observed, there are no attributes in the edges, each edge is defined by the pair of source and target ID of the entities it connects.
- Conferences included in the dataset are: AAAI, AAMAS, ACL, CHI, COLT, DCC, EC, FOCS, ICCV, ICDE, ICDM, ICML, ICSE, IJCAI, NIPS, SIGGRAPH, SIGIR, SIGMOD, UAI, and WWW.
- All papers must have a title and be connected to at least an author

Does it rely on external sources? - The dataset was originally built using papers and authors from the Microsoft Academic Graph⁴, however, there are no updates since its creation and therefore it does not rely on the Microsoft dataset for its use.

Arnab Sinha et al. (2015). "An Overview of Microsoft Academic Service (MAS) and Applications". In: *Proceedings of the 24th International Conference on World Wide Web. WWW '15 Companion*. Florence, Italy: Association for Computing Machinery, pp. 243–246. ISBN: 9781450334730. DOI: 10.1145/2740908.2742839. URL: <https://doi.org/10.1145/2740908.2742839>

Dataset splits or other samples available in the repository that contains this same data or the other way around - The dataset is split into 3 parts, train and test relationships and valid relationships. These splits can be used for benchmarking purposes in tasks in which this knowledge is used for graph data analysis. The dataset contains the following files: all_entity_info.txt, all_relation_info.txt, text.txt, train.txt and valid.txt. In the using the dataset section we give details on the format of each file.

How is it different from other versions? This dataset contains only papers from the mentioned top 20 conferences compared to its main source, Microsoft Academic dataset.

MOTIVATION (WHY?)

Why was the dataset created? - This dataset was created from the Microsoft Academic Graph (MAG) to support research on scholar metadata. In specific this dataset was created to serve as a benchmarking in order to reproduce results for known graph tasks such as link prediction, recommendation and question answering on scholar data. This dataset is curated to have papers from 20 selected high-ranked conferences and serves similar benchmarking purposes to other well-known datasets such as Freebase and DBLP.

Which task was it intended to solve? - The authors of the dataset have created it as a benchmarking for the tasks: knowledge graph completion and link prediction. There are no mentions of other graph data analysis tasks in which this dataset should not be used.

Has this data been used before? This data has been used in the first author PhD thesis in which they studied the use of knowledge graph embeddings (vector representation of knowledge graph information) for knowledge graph completion and link prediction. The paper's authors for each of the tasks follows:

- Hung Nghiep Tran and Atsuhiko Takasu (2020). "Multi-Partition Embedding Interaction with Block Term Format for Knowledge Graph Completion". In: *ECAI 2020 - 24th European Conference on Artificial Intelligence, 29 August-8 September 2020, Santiago de Compostela, Spain, August 29 - September 8, 2020 - Including 10th Conference on Prestigious Applications of Artificial Intelligence (PAIS 2020)*. Ed. by Giuseppe De Giacomo et al. Vol. 325. Frontiers in Artificial Intelligence and Applications. IOS Press, pp. 833–840. DOI: 10.3233/FAIA200173. URL: <https://doi.org/10.3233/FAIA200173>

⁴<https://dl.acm.org/doi/10.1145/2740908.2742839>

- Hung-Nghiep Tran and Atsuhiko Takasu (2022). “MEIM: Multi-partition Embedding Interaction Beyond Block Term Format for Efficient and Expressive Link Prediction”. In: *Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence, IJCAI 2022, Vienna, Austria, 23-29 July 2022*. Ed. by Luc De Raedt. ijcai.org, pp. 2262–2269. doi: 10.24963/IJCAI.2022/314. URL: <https://doi.org/10.24963/ijcai.2022/314>

If it has been used or is a sample, why was each of the entities chosen? This can be considered a sample of the Microsoft Academic Graph by considering only the top 20 conferences. There are samples available but the dataset is split into train and test sets to support the benchmarking of the tasks mentioned. There are no details available on how these entities were chosen.

Reasoning behind the modeling of the graph - The graph was modelled so it can be read also as a set of triples (similar to the RDF) in a way that it can be used in a similar way that other benchmarks of the area (e.g. Freebase dataset) can also be used.

DATA COLLECTION AND PRE-PROCESSING (WHEN? HOW?)

Source of data - The data was collected from the Microsoft Academic graph (MAG)⁵ although it is not mentioned which version or how the REST API was used.

Time-frame that the data was collected Not clearly mentioned but the dataset was first used by the authors in 2019 in the paper with reference below, a similar dataset was also used in a previous paper in 2016, however, it is not mentioned if it refers to the same dataset in the description.

- Hung Nghiep Tran and Atsuhiko Takasu (2019). “Exploring Scholarly Data by Semantic Query on Knowledge Graph Embedding Space”. In: *Digital Libraries for Open Knowledge - 23rd International Conference on Theory and Practice of Digital Libraries, TPD L 2019, Oslo, Norway, September 9-12, 2019, Proceedings*. Ed. by Antoine Doucet et al. Vol. 11799. Lecture Notes in Computer Science. Springer, pp. 154–162. doi: 10.1007/978-3-030-30760-8_14. URL: https://doi.org/10.1007/978-3-030-30760-8%5C_14
- Tin Huynh, Trac-Thuc Nguyen, and Hung Nghiep Tran (2016). “Exploiting Social Relations to Recommend Scientific Publications”. In: *Computational Social Networks - 5th International Conference, CSoNet 2016, Ho Chi Minh City, Vietnam, August 2-4, 2016, Proceedings*. Ed. by Hien T. Nguyen and Václav Snásel. Vol. 9795. Lecture Notes in Computer Science. Springer, pp. 182–192. doi: 10.1007/978-3-319-42345-6_16. URL: https://doi.org/10.1007/978-3-319-42345-6%5C_16

How the dataset was collected? “From the Microsoft Academic Graph dataset, we extracted high quality computer science papers published in top conferences between 1990 and 2010. The top conference list are based on the CORE ranking A* conferences. The data was cleaned by removing conferences with less than 300 publications and papers with less than 20 citations. The final list includes 20 top conferences: AAAI, AAMAS, ACL, CHI, COLT, DCC, EC, FOCS, ICCV, ICDE, ICDM, ICML, ICSE, IJCAI, NIPS, SIGGRAPH, SIGIR, SIGMOD, UAI, and WWW”. Information written by the dataset authors extracted from the dataset website⁶.

Is the dataset a sample? Dataset can be seen as a sample crawled from the Microsoft Academic Graph.

Is the raw data available? (if there is any) - Raw data is available through the Microsoft Academic Graph website, however, due to non-determined time frame by the dataset authors it might be difficult to reproduce.

⁵<https://www.microsoft.com/en-us/research/project/microsoft-academic-graph/>

⁶<https://www.kaggle.com/datasets/tranhungnghiep/kg20c-scholarly-knowledge-graph>

DISTRIBUTION AND MAINTENANCE (WHERE?)

License of the dataset	Attribution-NonCommercial 4.0 International (CC BY-NC 4.0) ⁷
Date of Distribution	Uploaded on October,2021 in Github and October,2025 in Kaggle
Dates of Modification	There were no modifications since its publication
Responsible for hosting the dataset	The dataset is being hosted in the public websites Github and Kaggle
Responsible for maintaining the dataset	Although not clearly mentioned, author Hung Nghiep Tran seems to be the responsible

Papers that use this dataset? The following papers use this dataset:

- Hung-Nghiep Tran and Atsuhiko Takasu (2022). “MEIM: Multi-partition Embedding Interaction Beyond Block Term Format for Efficient and Expressive Link Prediction”. In: *Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence, IJCAI 2022, Vienna, Austria, 23-29 July 2022*. Ed. by Luc De Raedt. ijcai.org, pp. 2262–2269. doi: 10.24963/IJCAI.2022/314. URL: <https://doi.org/10.24963/ijcai.2022/314>
- Hung Nghiep Tran and Atsuhiko Takasu (2020). “Multi-Partition Embedding Interaction with Block Term Format for Knowledge Graph Completion”. In: *ECAI 2020 - 24th European Conference on Artificial Intelligence, 29 August-8 September 2020, Santiago de Compostela, Spain, August 29 - September 8, 2020 - Including 10th Conference on Prestigious Applications of Artificial Intelligence (PAIS 2020)*. Ed. by Giuseppe De Giacomo et al. Vol. 325. Frontiers in Artificial Intelligence and Applications. IOS Press, pp. 833–840. doi: 10.3233/FAIA200173. URL: <https://doi.org/10.3233/FAIA200173>
- Hung Nghiep Tran and Atsuhiko Takasu (2019). “Exploring Scholarly Data by Semantic Query on Knowledge Graph Embedding Space”. In: *Digital Libraries for Open Knowledge - 23rd International Conference on Theory and Practice of Digital Libraries, TPDL 2019, Oslo, Norway, September 9-12, 2019, Proceedings*. Ed. by Antoine Doucet et al. Vol. 11799. Lecture Notes in Computer Science. Springer, pp. 154–162. doi: 10.1007/978-3-030-30760-8_14. URL: https://doi.org/10.1007/978-3-030-30760-8%5C_14

How can a third party contribute to this dataset? - There are no instructions from the authors on how to contribute to this dataset.

USING THE DATASET AND LEGAL/ETHICAL ASPECTS (HOW?)

Was this dataset approved by an ethical review? - This dataset did not go through an ethical review process as it is based on a previous public dataset (Microsoft Academic Graph).

Privacy issues - The data was no anonymized, all data in this graph belongs to publicly available data on research papers.

Ethical issues - It is unknown if using this graph for learning tasks could lead to potential advantage or disadvantage to specific groups. However, the main idea of the dataset is that it is a curated citation graph from the top ranked conferences, this means that there are probably more authors and affiliations which frequently publish in such conferences. According to statistics, it is more common that authors and affiliations from the global north participate⁸ in such conferences which causes a disbalance in this dataset and could potentially lead to disadvantages to authors and affiliations from the global south, although this has not been verified in any previous work using this dataset.

Does it contain inappropriate information? Or any potentially inappropriate data? -It does not contain inappropriate information.

⁸<https://www.nature.com/articles/d41586-023-03901-x>

Does it comply with a data protection regulation law? - Although not mentioned, given that the dataset was created with publicly available data it does not indicate any possible issues with not complying with data protection regulation law.

How to use the dataset? This dataset contains the following files: all_entity_info.txt, all_relation_info.txt, train.txt, valid.txt, test.txt. We go through each one of the files and explain its format.

all_entity_info.txt

This file contains information about the entities in this graph, that is the node Labels: Paper, Author, Affiliation, Conference, Domain. The file can be read as a csv file in which the separator is a \t space and contains 3 columns: ID, Name, type in which the type refers to the label.

Preview of the dataset:

```
id name type
7C7CAEED On rank correlation in information retrieval evaluation paper
7AEE29E3 The Voting Model for People Search paper
7D68490B Document clustering with committees paper
7A488256 A comparison of indexing techniques for Japanese text retrieval paper
7D5CD2DF Feature selection for ranking paper
```

all_relation_info.txt

Contains the list of relationships/edge labels in this dataset

train.txt-valid.txt-test.txt

All these files although different splits (described in Data Collection section), have the same format. These files contain the relationships between the nodes of the all_entity_info.txt also in text files which could be read as a csv. The format of these files are: entity_1_ID relationship entity_2_ID and there are **no headers** in these files. The entity_1_ID and entity_2_ID are IDs from the first column in the all_entity_info.txt file. It follows a preview of the format:

```
7B50CE3E author_in_affiliation 4E13CA1C
78F30396 author_in_affiliation 4FF45383
80582427 paper_in_domain 0247DE90
8125EA5C author_in_affiliation 0477FFD3
7DEDE08C paper_in_domain 05758F58
```

Using the splits: If the task in which you are using this dataset for, does not need the mentioned splits (train and test), it is possible to include all relationships in a single graph (add all relationship files, train.txt, test.txt and valid.txt) so that citations of a paper can be retrieved or other graph metrics (centrality measures, for example) can be with the full set of relationships.